

Arithmetic Logic Unit (ALU)**INTEGER DIVISION**Sem: 4th

Section: COA-CSE-G1 & COA-CSCE-G1

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Lecturer Summary

1. Understanding the process of integer division
2. Restoring Division Algorithm
3. Non-restoring Division Algorithm

1 Understand process of integer division**Division using successive subtraction**

- Implemented on computer systems by repeatedly subtracting the divisor from the dividend
- Counting the number of times that the divisor can be subtracted from the dividend before the dividend becomes smaller than the divisor.
- **Division 17 with 5:** Subtract repeatedly from 17, getting 12, 7, and 2 as intermediate results. The quotient, 3, is the number of subtractions that had to be performed before the intermediate result became less than the dividend.
- **17=10001 and 5=101, Now $17 \div 5$**

$$\begin{array}{r}
 \text{Divisor} \longrightarrow 1 \ 0 \ 1 \quad \left| \begin{array}{r}
 \begin{array}{r}
 0 \ 1 \ 1 \longleftarrow \text{Quotient} \\
 1 \ 0 \ 0 \ 0 \ 1 \longleftarrow \text{Dividend} \\
 \underline{0 \ 0 \ 0} \\
 1 \ 0 \ 0 \ 0 \\
 \underline{1 \ 0 \ 1} \\
 1 \ 1 \ 1 \\
 \underline{1 \ 0 \ 1} \\
 1 \ 0
 \end{array}
 \end{array}
 \right.
 \end{array}$$

- **Disadvantages:** This method takes too long time.
 - For example, 2^{31} (one of the larger numbers representable in 32-bit unsigned integers) divided by 2 is 2^{30} , meaning that 2^{30} subtractions would have to be done to perform this division by repeated subtraction.
 - On a system operating at 1 GHz, this would take approximately 1s, far longer than any other arithmetic operation.

2 Restoring Division Algorithm

Step-1: Load the n-bit positive divisor into register M, Load the n-bit positive dividend into register Q. Register A is set to zero.

Step-2: Do the following n times.

- A, Q << 1:** Shift A and Q left one binary position. Leave LSB of A, Q blank (q_0).
- A = A - M:** Subtract M from A, and place the answer back in A.
- If the **sign** of A is 1, set $q_0 = 0$ and do **A = A + M**: add M back to A (That is restore A)
Else set $q_0 = 1$

Step-3: After the division is complete, the n-bit quotient is in register Q and the remainder is in register A.

N.B. The extra bit position at the left end of A and M accommodates the sign bit during subtractions.

Example-1: Divide 8 by 3 using restoring division method.

Solution: Dividend=Q=8=1000, Divisor=M=3=11

As length of dividend is 4 bit, So length of A and M should be (4+1)=5bit.

So, A=00000 and M=00011 $M^{2s}=11101$

Steps	Sub-step	Step/Action	Accumulator (A)	Dividend (Q)	Divisor/Remarks (M)
0	0	Initial values	00000	1000	00011
First Cycle	a)	Shift left A,Q	00001	000	
	b)	Perform $A = A - M$ (A-Column)	11110	000	$A = A - M = A + (-M) = A + M^{2s}$ $A = 00001$ $M^{2s} = 11101$ Addition = 11110
	c)	Now A is -ve, set $q_0 = 0$ (Q-column) and perform $A = A + M$ (A-Column)	00001	0000	$A = A + M$ $A = 11110$ $M = 00011$ Addition = 00001
Second Cycle	a)	Shift left A,Q	00010	000	
	b)	Perform $A = A - M$ (A-Column)	11111	000	$A = A - M = A + (-M) = A + M^{2s}$ $A = 00010$ $M^{2s} = 11101$ Addition = 11111
	c)	Now A is -ve, set $q_0 = 0$ (Q-column) and perform $A = A + M$ (A-Column)	00010	0000	$A = A + M$ $A = 11111$ $M = 00011$ Addition = 00010
Third Cycle	a)	Shift left A,Q	00100	000	
	b)	Perform $A = A - M$ (A-Column)	00001	000	$A = A - M = A + (-M) = A + M^{2s}$ $A = 00100$ $M^{2s} = 11101$ Addition = 00001
	c)	Now A is +ve, set $q_0 = 1$ (Q-column)	00001	0001	
Fourth Cycle	a)	Shift left A,Q	00010	001	
	b)	Perform $A = A - M$ (A-Column)	11111	001	$A = A - M = A + (-M) = A + M^{2s}$ $A = 00010$ $M^{2s} = 11101$ Addition = 11111
	c)	Now A is -ve, set $q_0 = 0$ (Q-column) and perform $A = A + M$ (A-Column)	00010	0010	$A = A + M$ $A = 11111$ $M = 00011$ Addition = 00010

So Quotient is the contains of register Q=0010=2

And remainder is the contains of register A=00010=2

Example-2

Divide 163 by 11 using restoring division method.

Iteration	Sub-step	Step/Action	Accumulator (A)	Dividend (Q)	Divisor/Remarks (M)
0	0	Initial values	00000000	10100011	00001011
1	a)	Shift left A,Q	00000001	0100011	
	b)	Perform $A = A - M$ (Only changes will be in A column, column Q remains unchanged)	11110110	0100011	$A = A - M = A + (-M) = A + M^{2s}$ $A = 00000000$ $M^{2s} = 11110101$ Addition = 11110110
	c)	Now A is -ve (as sign bit of A is 1), So set $Q_0 = 0$ (Only changes will be in Q column, column A remains unchanged)	00000001	01000110	$A = A + M$ $A = 11110110$ $M = 00001011$ Addition = 11110110
2	a)	Shift left A,Q	00000010	1000110	
	b)	Perform $A = A - M$	11111010	1000110	
			Rest	Do	yourself

 $M^{2s} = 2$'s complement of M**3 Non-restoring Division Algorithm****Step-1:** Load the n-bit positive divisor into register M, Load the n-bit positive dividend into register Q. Register A is set to zero.**Step-2:** Do the following n times.

a) If the sign. of A is 1, shift A and Q left one position ($A, Q \ll 1$) and add M to A ($A = A + M$)
else shift A, Q left and subtract M from A ($A = A - M$)

b) If the **sign** of A is 1, **set $q_0 = 0$** **else set $q_0 = 1$**

Step-3: After the division is complete, the n-bit quotient is in register Q and the remainder is in register A.**Step-4:** if the sign of A is 1, add M to A ($A = A + M$).

Example-1: Divide 8 by 3 using non-restoring division method.

Solution: Dividend=Q=8=1000, Divisor=M=3=11

As length of dividend is 4 bit, take A and M should be (4+1)=5bit.

So, A=00000 and M=00011 $M^{2s}=11101$

Steps	Sub-step	Step/Action	Accumulator (A)	Dividend (Q)	Divisor/Remarks (M)
0	0	Initial values	00000	1000	00011
First Cycle	a)	As sign of A is 0, so Shift left A, Q and do	00001	000_	
		Subtract (A=A-M)	11110	000_	$A=A-M=A+(-M)=A+M^{2s}$ A = 00001 M^{2s} = 11101 Addition = 11110
	b)	Set $q_0=0$ as sign of A is 1	11110	000 <u>0</u>	
Second Cycle	a)	As sign of A is 1, so Shift left A, Q and do	11100	000_	
		Add (A=A+M)	11111	000_	A=A+M A = 11100 M = 00011 Addition = 11111
	b)	Set $q_0=0$ as sign of A is 1	11111	000 <u>0</u>	
Third Cycle	a)	As sign of A is 1, so Shift left A, Q and do	11110	000_	
		Add (A=A+M)	00001	000_	A=A+M A = 11110 M = 00011 Addition = 00001
	b)	Set $q_0=1$ as sign of A is 0	00001	000 <u>1</u>	
Fourth Cycle	a)	As sign of A is 0, so Shift left A, Q and do	00010	001_	
		Subtract (A=A-M)	11111	001 <u>0</u>	$A=A-M=A+(-M)=A+M^{2s}$ A = 00010 M^{2s} = 11101 Addition = 11111
	b)	Set $q_0=0$ as sign of A is 1	11111	001 <u>0</u>	

As the sign of A is 1, so do $A=A+M=11111+00011=00010$ =Remainder

So Quotient is the contains of register Q=0010=2